HarvestWise AI — Phase 3 Evaluation Report

Harvest Timing & Market Option Decision Support Simulator
Generated: 2026-10-29 14:01:26 | Environment: Local Offline Verified

1. System Architecture & Objectives

HarvestWise AI is a transparent, rule-based decision support simulator engineered for smallholder horticulture farmers. Phase 3 focuses on evaluating trade-offs between immediate harvest, speculative waiting for price recovery, and distant market arbitrage. The system explicitly incorporates post-harvest spoilage degradation, weather adversity factors, cumulative holding costs, and transport transit friction without double-counting operational expenses.

2. Reproducible Experiments & Baseline Auditing

Documented Baseline Formulation: Naive smallholder planning evaluates potential revenue as Quantity × Spot Price - Transport Cost, wholly neglecting storage fees and in-transit/holding decay. HarvestWise AI rigorously benchmarks every decision against this naive baseline.

ID	Scenario	Weather	Naive Base (■)	Now Net (■)	Wait Net (■)	Alt Net (■)	Recommendation
A	Mature Crop with Sta	Low	■28,800	■28,086	■27,672	■25,715	Harvest Now (Nearb
B	Price Rise with Rain	High	■22,600	■19,977	■26,107	■17,922	Wait for Better Pr
C	Higher-Price Alterna	Moderate	■23,800	■22,920	■23,885	■30,789	Alternate / Distan

3. Prototype Error Analysis & Limitations

1. Baseline Overestimation: In high humidity and wet seasons (Scenario B), the naive baseline overestimates net farmer earnings by upwards of ■3,000+ because it ignores storage rent and accelerated rotting.
2. Spoilage Sensitivity: The optimal decision is highly non-linear. Spoilage rates above 18% rapidly negate any modest market price increase.
3. Transport Transit Risk: Transit durations exceeding 6 hours in non-refrigerated transport cause transit decay that frequently cancels distant price arbitrage.

4. Responsible AI, Multilingual Support & Offline Protocol

• **Human-in-the-Loop:** The prototype provides non-prescriptive advisory decision support. Crop cuts and marketing remain with the farmer.
• **Immaturity Safeguard:** Automatically flags crops below 65% maturity to avoid harvest losses.
• **Multilingual & Privacy:** Full English and Tamil bilingual support. Zero personally identifiable farmer data is stored.
• **Offline First:** Pure local computation. Fallback regional heuristics allow operation even when data feeds are offline.

5. Automated Test Verification Audit

Automated regression test suites (tests/test_simulator.py and tests/test_validation.py) verify non-double counting of expenses, edge cases (zero quantity, extreme spoilage, negative prices), heuristic bounds, and decision flips. All tests pass with zero errors.